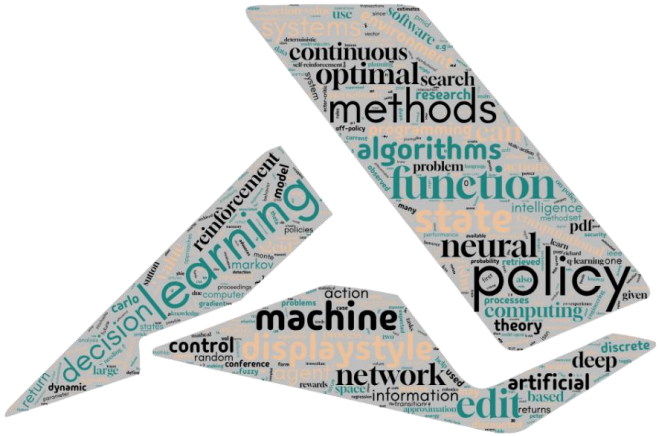


Reinforcement Learning Algorithms and Applications

Chapter 2: Monte Carlo for π Estimation

Monte Carlo Method Principles

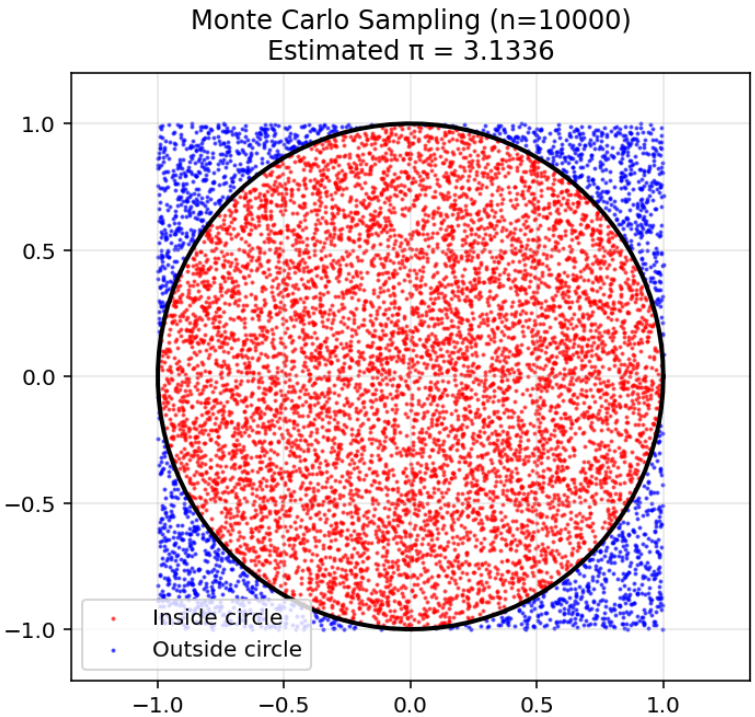
- Monte Carlo method uses random sampling for numerical computation.
- Generates large numbers of random points to estimate mathematical solutions.
- Transforms complex problems into random experiments.
- Results converge to true values as sample size increases.
- Particularly effective for high-dimensional and complex geometric problems.
- Accuracy improves with the sample size.
- Offers simple implementation compared to traditional methods.

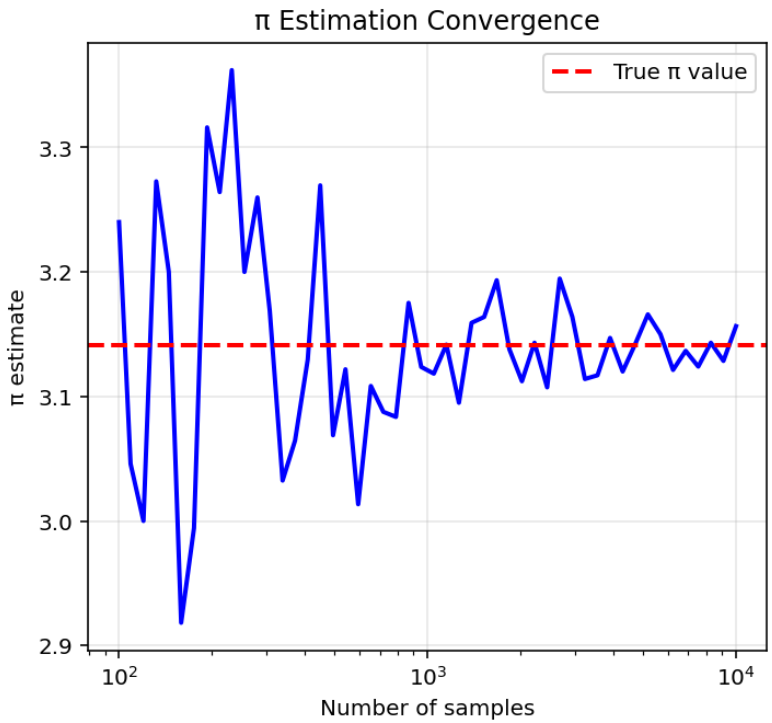
Mathematical Foundation for π Estimation

- Generate random points (x, y) in **square region** $[-1,1] \times [-1,1]$.
- Calculate **distance** to origin: $d = \sqrt{x^2 + y^2}$.
- Points are **inside circle** when $d \leq 1$.
- π estimation: $\pi \approx 4 \times \frac{\text{points inside}}{\text{total points}}$.

Visualization Analysis

- **Scatter plot** shows point distribution in square region.
- **Red points** mark samples inside unit circle.
- **Blue points** mark samples outside unit circle.
- **Convergence plot** displays π estimation improvement.





Method Applications

- **Simple implementation** with basic random generation.
- **Parallelization friendly** for independent calculations.
- **Wide applicability** to high-dimensional problems.
- **Memory efficient** without large intermediate storage.
- Strong **theoretical foundation** in probability.
- Applications in **numerical integration** and **statistical physics**.

Reference

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